

R: A DATA-ANALYSIS And STATISTICAL-COMPUTING Tool

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R is a statistical-computing environment that consists of a language along with a run-time environment with graphics, a debugger, access to certain system functions and the ability to run programs stored in script files. Implementation of R is heavily influenced by two programming languages: S and Scheme. R has inherited strong object-oriented features from S language. The underlying implementation and semantics of the software is inspired from Scheme.

R provides a wide range of statistical techniques such as linear and non-linear modelling, classical statistical tests, time-series analysis, classification clustering and more. Communication engineers make use of R for various signal-processing techniques such as filter design and processing of electrocardiogram (ECG) signals.

The software was initially created by Ross Ihaka and Robert Gentleman at University of Auckland, New Zealand, in 1993, and is currently being developed by R Development Core Team. The latest stable release of R version 3.2.3 is bundled in DVD accompanying this month's EFY Plus.

Why I should go for R

With a lot of software packages available for data analysis and statistical computing, one might question the relevance of yet another package for carrying out the same.

Benefits of R being a language. R is not an easy-to-learn language by itself. But definitely it has a

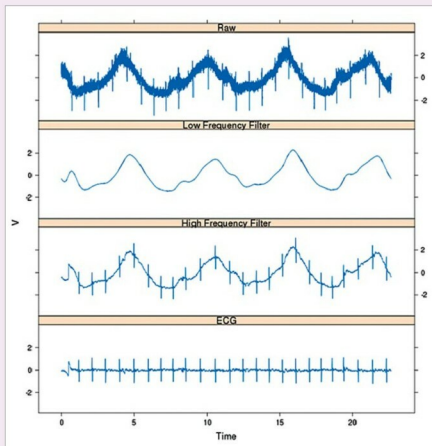


Fig. 1: Reproducing ECG signals from raw data (Courtesy: <http://biostatmatt.com>)

lot of advantages over other data-analysis tools, if you could master it. R, being a highly-interactive language, allows the programmer to experiment and explore new areas and new functionalities. This would not be even possible if the data-analysis tool was not a language. The script can be re-run any time, on any machine.

Cutting-edge analytics. A powerful analytics software should be capable of accepting data in various formats, manipulating and convert-

And so they called it R

There are two reasons behind the naming of this software. First, it comes from the names of its two creators, Ross Ihaka and Robert Gentleman.

Second, the name R acknowledges its inspiration from software S.

ing it to traditional and modern statistical models. R does all of these. Various manipulations like transforms, merge, aggregations and so on are carried out on accepted data